

Supporting the Long Now: Building Technology for Tangible Interaction in a Decade-Long Exhibition

Andy Sheen*

Interaction Research Studio
Northumbria University
London, United Kingdom
andrew.sheen@northumbria.ac.uk

Michail Vanis*

Interaction Research Studio
Northumbria University
London, United Kingdom
mike.vanis@northumbria.ac.uk



Figure 1: Farm-Free Protein Machine (left), Highpoint (middle) and Better Life (right), three devices developed for the exhibition.

Abstract

Designing interactive systems for long-term deployment in public museums introduces unique challenges that are not often addressed in research-through-design or domestic HCI contexts. In this paper, we document the technical development of three bespoke interactive devices embedded within a decade-long exhibition at a museum. Our work builds on the concept of research products, adapting their core principles to a fixed, high-traffic museum context. We describe the hardware and software strategies used to build robust, maintainable systems capable of evolving over time without reliance on external services. A year into the decade-long exhibition, we reflect

on the role of maintenance, repair, and care in sustaining these systems throughout the lifetime of the exhibition, and contribute practical strategies for supporting longevity in tangible systems intended to live in the "long now".

CCS Concepts

• **Hardware** → *Hardware reliability*; • **Computer systems organization** → *Architectures*; • **Applied computing** → *Electronics*; • **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction design**.

*Both authors contributed equally to this research.



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Keywords

Interactive Exhibits, Museum Technology, Hardware design, Design for Longevity, Maintenance and repair, Public installations, embedded systems

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1 Introduction

We worked with a public museum focused on the history and meanings of domestic life. Its displays include a chronological sequence of period-room environments, staged with objects and stories that connect everyday home life to local communities. In 2024, the museum launched a refreshed permanent display intended to broaden the range of voices represented across the rooms. The concluding room departs from the historical sequence, imagining a near-future domestic interior in a converted flat set in 2049.

Our team was invited to design and fabricate this future domestic space. The brief was to engage the public in imagining how everyday home life might evolve in the coming decades. The museum itself operates six days a week throughout the year, with the exhibition scheduled to run for 10 years. Now a year in its 10-year lifespan, the room presents an evolving set of interactive artefacts that can be updated and reconfigured over time.

We align with research-through-design (RtD) [18] work that involves the design of research products [40], with interactive devices and systems deployed with participants over weeks or months, to support long-term engagement and inquiry [21–24, 33]. Research products are typically used for an extended period in domestic environments. In contrast, the exhibition described in this paper posed a markedly different set of technical design challenges. It is located in a high-traffic public museum, requiring all devices to operate continuously, withstand significant physical interaction from all age groups, and remain maintainable and extensible throughout the exhibition's decade-long lifespan.

Research products have been extensively explored in domestic contexts, with their deployment typically spanning weeks or months [7, 40]. Meanwhile, museum HCI primarily examines visitor engagement, learning, and interaction design, with less literature on the infrastructural challenges of sustaining interactive systems in public settings for long-periods. There is an opportunity in investigating how the core qualities of research products – inquiry driven, finish, fit, and independence – translate to fixed, high-traffic environments where devices must operate continuously, withstand heavy use, and remain maintainable without the original designers' ongoing involvement. In this paper, we ask: what hardware and software strategies support the development and sustained deployment of interactive research products in long-term exhibitions? We address this question through a reflective account of designing, deploying, and maintaining three interactive devices over the first 12 months of a decade-long exhibition. By foregrounding maintenance, repair and iteration as ongoing creative and technical practices, this work speaks to broader questions within the Creativity and Cognition field around the temporality of designed artifacts [43] and the sustained creative labour involved in keeping interactive systems alive over time [39].

Alongside the technical composition of the three interactive artefacts discussed in this paper, we contribute a set of practical

strategies derived from our experience of deploying and maintaining these devices over the first 12 months of a decade-long exhibition. These strategies combine anticipated hardware decisions and technical approaches that proved effective for long-term public installations, alongside insights that emerged through ongoing maintenance, care, and repair. Our experiences foreground how these strategies became meaningful in practice, and how additional, unanticipated strategies became critical to sustaining the devices over time.

2 Related Work

In this section we discuss existing work that is relevant to the technical development of *Better Life*, *Farm-Free Protein Machine*, and *Highpoint*. We choose not to discuss museum studies work extensively, as this primarily focuses on the design process, visitor experience, and curatorial decisions of the overall exhibition. Given the focus of the paper is around robustness and repairability, we explore how research products relate to our work, and how relevant the key qualities of research products are when developing tangible interactives for this museum context. Additionally, we draw connections with the concept of the *Long Now* [8], and ideas of building robustness and inspiring repair.

HCI has long played a significant role in the design of interactive museum exhibits, with an abundance of prior work that focuses on aspects of visitor engagement, visitor learning outcomes, and interactive storytelling, using tangible artifacts and hybrid physical-digital systems [5, 12, 13, 26, 36, 42]. These works have contributed to the understanding of how technology can be used to enhance museum experiences, highlighting how interactive exhibits can help reinforce learning, knowledge transfer and cultural heritage. While museum HCI literature typically evaluates interactive exhibits through visitor studies and feedback, it tends to not address the long-term practical and infrastructural challenges of deploying interactive systems in museum settings, such as breakdown, repair, and maintenance.

In contrast, this paper more closely relates to HCI research on robust, tangible, and repairable systems, where design decisions are foregrounded as a means of supporting longevity, maintainability, and resilience in real-world deployments. By presenting design strategies rather than visitor evaluation, this paper adds to existing museum HCI by focusing on how interactive museum systems can be designed and maintained to endure high-footfall exhibitions over extended periods of time.

2.1 Research Products

Our work builds on the growing body of literature surrounding research products; artefacts designed with a high degree of finish, intended to live in the world and provoke the reflection of participants [40]. Research products embody four key qualities: inquiry-driven, finish, fit, and independence. They have been widely used in the context of home-based, long-term studies in HCI research [15, 33, 41, 46]. Prior work with research products has involved designing interactive systems that are intended to be used in domestic settings, used by individuals or households over periods ranging from weeks to years [7]. In these contexts, the independence of a

research product refers to its autonomy from the researcher and its ability to operate without technical intervention.

With the interactives presented in this paper, we carry these qualities into a different domain: a public museum. Our devices are similarly inquiry-driven, probing how people imagine and relate to future domestic life. However, their finish must reflect not the aesthetics of a research prototype, but the credibility of a speculative consumer product. Their fit is speculative in a different sense than domestic research products, since they anticipate and perform a fictional yet plausible near-future domestic setting. The devices suggest how technology might operate, appear, and be interacted with in 2049, and as such, balance familiarity with novelty.

Beyond their setting, research products differ from prototypes because they represent "what it is as opposed to what it might become" [40] whereas prototypes embody only aspects of an envisioned complete product [27]. Our museum devices are a distinct category: like prototypes, they only simulate the products they represent. Yet unlike prototypes, they do not serve as interim stages on the way to complete products. Instead, like research products, they are finished realisations of what they are meant to be – if it is understood that what they are meant to be is simulated future products.

2.2 Breaking the Shell

Many research products are deployed in homes, and are often physically mobile or embedded in varying physical contexts [10, 16, 32, 35, 50]. Research product devices are comprised of an enclosure, with all required technology embedded within it. However, devices in a museum exhibition are fixed in place. They are designed to live in a single, carefully staged environment within a room in the exhibition. This fixed state presents an opportunity to externalise certain technical components: rather than embedding all electronics within the device, they can be distributed in cupboards, plinths, and hidden nooks directly underneath the devices. This strategy resonates with Chatting et al.'s Pace Layers in Research Products [11], adapted from Brand's Pace Layering diagram [9]. Breaking some of the hardware and software out of the shell of the object allows better physical access and therefore easier maintenance and longevity.

In Chatting et al.'s Pace Layering, "Stuff" relies on "Infrastructure" to operate; remote servers passing data back and forth, WiFi routers connecting devices to the internet, or the power grid supplying electricity. Actively reducing the "Stuff's" reliance on the Infrastructures layer makes objects more resilient and long-lasting. Over extended deployments, decoupling functionality from external infrastructure protects against servers going offline, internet failures, and remote technologies becoming obsolete. In our work, we apply this principle by opting for cloudless, offline processes and avoid reliance on external servers and APIs that may disappear or change within the exhibition's ten-year lifespan.

2.3 Maintenance and Care

We also draw inspiration from *The Clock of the Long Now*, and the notion of the "Long Now", as formulated by Brand and Eno [8]. The clock is designed not only to function over millennia, but also to provoke interest in its care and upkeep. Similarly, our devices

are not sealed systems that are never intended to be opened. They are meant to inspire maintenance, inviting future curators and researchers to care for the objects as the exhibition evolves. This echoes discussions around maintenance and care in HCI, where systems are deliberately constructed to foreground reparability, adaptation, and technical legibility [28, 30].

2.4 Smoke-and-mirrors

Finally, we are influenced by Andersen's concept of magic machines: speculative artefacts that favour experience, narrative, and symbolic interaction over functional complexity [4]. With our devices, we aim to evoke the same curiosity and reflection, deploying "smoke-and-mirrors" (illusions that hide the real mechanics of a magic trick) to aid the fiction that the object lives within. Technically this manifests in technology that favours an experience over making something work for real. For example, one of our devices uses a visual illusion and a concealed chamber to appear as if it is dispensing food, rather than extruding food, which would be both impractical and difficult in the context of the exhibition.

3 Exhibit Overview

In this section, we give a brief overview of the overall exhibition our three artefacts are located in. We present the devices' concept and the room's intended experience.

3.1 Exhibit Concept

The exhibition described in this paper presents a speculative home set in the year 2049. Instead of aiming to predict a singular future, the room uses the trope of a multi-generational household to explore socio-political, environmental and technological trajectories. Partial narratives of the room's inhabitants are dotted around the room in the form of possessions, memorabilia, devices, and exhibition text, inviting visitors to reflect on how homes might adapt to shifting climates, economies, and values.

3.2 Interactive Devices

The exhibit includes, amongst many other objects, three bespoke devices that the public can interact with. Their concept is detailed here for context. *Better Life* (Figure 2) is a digital photo frame that cycles through family photographs. A dial labelled "Normal", "Better" and "Even Better" allows the user to enhance each photo with AI, with each level making the subject more impressive and outlandish. *Highpoint* (Figure 7) is a VR-like sensory stimulation headset that produces enhanced hallucinations of real memories from the family. Pulsating audio and LEDs positioned around the wearer's eyes contribute to the experience, inspired by EMDR therapy [45]. The *Farm-Free Protein Machine* (Figure 5) is a desktop protein fermenter designed for home food production, reflecting on post-industrial food practices and the family's concerns for the environment. Users can press buttons on the device to view usage logs, flavour preferences, and "dispense" the protein substance that the machine creates.

3.3 Intended Experience

The exhibition is fully open, allowing visitors to touch and interact with everything in the space. Designed with accessibility across a

wide age group, the room allows for multiple visitor experiences. For example, some visitors may purely interact with the physical devices and objects in the space, while others are more focussed on the text displayed in panels. Providing multiple entry points to experience the room enables a wider range of people to engage with the exhibit.

3.4 Exhibition Lifetime

The exhibition is intended to stay open for 10 years. Therefore, the room allows for expansion and change, with the intention of implementing future devices into the room. Already within the first year of its existence, new devices and objects have made an appearance. In the future, the possibility is open for us to completely redesign the room and the future within it, if we would like to.

4 Technical Implementation of Devices

In this section, we describe the technical implementation of the three interactive devices. Rather than focusing on the conceptual motivations behind each design, we detail the hardware and software components used in their construction.

4.1 Better Life Picture Frame



Figure 2: Better Life picture frame.

Better Life is an interactive digital photo frame that cycles through a carousel of images. Visitors can alter the sequence of images through physical interaction, engaging with a speculative concept that explores how AI-enhanced photography might be used in the future to reshape or distort personal memories.

The hardware (Figure 3) consists of an Amazon Fire TV Stick connected to a 2K IPS LCD display. A custom Unity application [47] running on the Fire TV Stick displays the images stored within the frame. The application includes video transitions between images, generated using TouchDesigner [14], which produce the illusion of

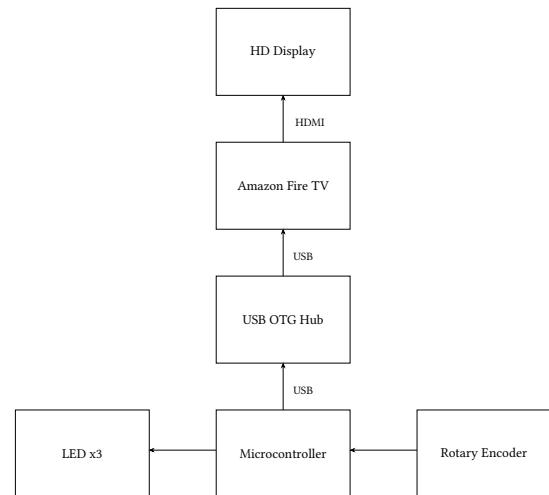


Figure 3: Block Diagram of the system architecture for Better Life.



Figure 4: A screenshot of the image transition sequence on Better Life.

one image transforming into another (Figure 4). User interaction is handled by a microcontroller connected to the Fire TV Stick through USB On-The-Go (OTG). The microcontroller functions as a Human Interface Device (HID), and includes a rotary encoder and three LEDs, allowing users to navigate the image carousel by turning the encoder, communicating user input to the Unity software application.

4.2 Farm Free Protein Machine

The Farm Free Protein Machine is a speculative food appliance designed to simulate a futuristic algae growth and fermentation process, creating the illusion that a protein-enriched substance is synthesized and distributed for household consumption.



Figure 5: Farm Free Protein Machine.



Figure 7: Highpoint memory intensification system.

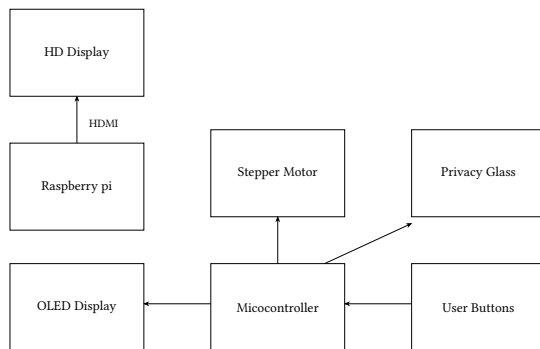


Figure 6: Block Diagram of the system architecture for the Farm Free Protein Machine.

The device is powered by a Raspberry Pi 4 connected to a 2K AMOLED (Active Matrix Organic Light Emitting Diode) display that continuously plays a prerecorded video using VideoLANs VLC Media player [48], which is controlled by a custom Python script. An additional 2.42-inch 128-by-64-pixel OLED (Organic Light Emitting Diode) screen, along with four illuminated push buttons and a stepper motor, are controlled by a microcontroller connected to the Raspberry Pi via USB (Figure 6).

The OLED screen displays a menu that can be browsed with two push buttons. Another button labelled “dispense” initiates the dispensing sequence: a 3D-printed barberpole illusion [49] rotates to simulate protein matter falling from the machine. To enhance the illusion, a section of switchable privacy glass conceals the internal mechanism while inactive.

4.3 Highpoint Memory Intensification System

The Highpoint Memory Intensification System is a speculative device envisioning an evolution of virtual reality headsets. It integrates EMDR (Eye Movement Desensitization and Reprocessing) [45] and brainwave entrainment [29], using synchronized visual and auditory stimuli to promote memory plasticity, transcendental states, and mimic the psychedelic effects of hallucinogens.

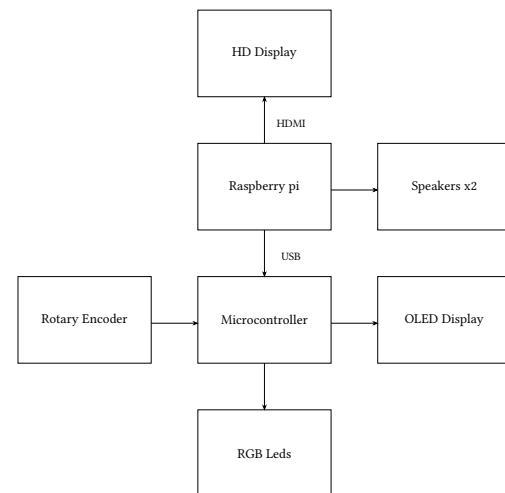


Figure 8: Block Diagram of the system architecture for the Highpoint Memory Intensification System.

The device is built around a Raspberry Pi 4 connected to a narrow letterbox-style display (Figure 8), which presents custom visual content created in TouchDesigner. The Raspberry Pi runs a Python

script that controls VLC Media Player [48] to manage video playback. Embedded in the device's handheld grips are micro-speakers that deliver synchronized vibration-like audio cues.

User interaction is handled by a microcontroller connected to the Pi via USB. This microcontroller emulates an HID keyboard device and provides simple control commands to switch between content. It also hosts a small OLED screen and a rotary encoder, forming a menu interface that allows users to select different visual and audio programs. Addressable RGB LEDs are used to generate dynamic lighting patterns synchronized with the video content.

5 Practical Strategies Deployed

In this section, we describe the key strategies applied during initial development of the three interactive devices. From the outset, we knew that the interactives were intended to remain in use for at least ten years, therefore we established a shared set of technical guidelines to support robustness and longevity across all devices in the exhibition. These guidelines provided a framework for hardware and software development, helping to ensure consistency in technical approach across the exhibition. This was particularly important given the short overall development timeline, which required the conceptual and physical design of the devices to proceed in parallel with technical development. The strategies presented here emerged through the prototyping of technologies used in the devices.

5.1 Keep It Simple

Throughout the development of the interactive devices, we followed the principle of simplifying both hardware and software wherever possible. In *Better Life* and *Highpoint*, the microcontrollers were configured to act as USB Human Interface Devices (HID), sending simple keystroke-style signals to the host system, much like a USB keyboard. This approach allowed plug-and-play compatibility with platforms like the Fire TV Stick or Raspberry Pi, without requiring custom drivers, complex integration, or hard-to-debug serial protocols.

Because the control interfaces were deliberately minimal, we found that one-way communication from the microcontroller to the content player was sufficient. There was no need for bi-directional communication or feedback from the host to the microcontroller. This decision significantly reduced system complexity and potential failure points. Additionally, it made testing and development more efficient, as the input signals could easily be emulated using a standard keyboard or mouse. This allowed the content layer to be prototyped, debugged, and refined without requiring the physical hardware to be connected.

To further support maintainability, we kept microcontroller code simple and offloaded most application logic to higher-level environments such as Python or Unity running on the host device. Microcontrollers were tasked solely with low-level hardware interaction (e.g., reading inputs, controlling LEDs), while all content logic and state management remained on the host. This separation allowed for clearer division of responsibilities, easier debugging, and more agile iteration across the team.

5.2 Cloudless

Whilst evaluating similar devices, we noticed that some prior research products relied on online services [20], custom servers [6], or third-party APIs [19]. It would be more convenient for *Highpoint*, *Better Life* and *Farm-Free Protein Machine* to access data from cloud services, as this would allow for remote updates to content. For example, *Better Life* could download images displayed on the screen from an online storage service, which we could change remotely. Similarly with *Highpoint*, videos could be downloaded and updated remotely. However, we opted for this content to exist on removable USB storage plugged directly into the devices. This ensured that if the Internet connection failed, the software would still have access to the content, preventing disruptions to the visitor experience.

During our testing, we found that some of these devices occasionally update their operating systems without warning. The Amazon Fire TV Stick flashed warnings about completing updates, which were overlaid on top of our content, which broke the illusion of the design fiction. We ran long-term tests to find out whether the Fire TV Stick would continue to display these update messages without being connected to the internet. Luckily, it worked perfectly without internet and the update alerts disappeared.

It became clear from the start that ideas for devices in the room would require interconnectivity to create more interesting narratives within the room. Themes around energy blackouts and pandemics would require room scenes that needed device choreography to enrich the world building. To make this possible, we installed Home Assistant in the room: an open-source framework for local home automation [17]. With this system, we prototyped choreographing events in which a device would change mode, a smart bulb would change color, and a speaker would play an audio file. This allowed for future updates where we change whole scenes within the room, and the three devices respond accordingly, without relying on a cloud. We decided to install Home Assistant Green [38], a plug-and-play device that runs Home Assistant, avoiding the need to manage a distribution on our own hardware.

5.3 Video production and playback on integrated screens

Reliable audiovisual content was an early sticking point during the design of the room. All three of the devices play video content in different forms. The Protein Machine shows how algae would be transformed into protein. *HighPoint* displays distorted videos inspired by the room's inhabitants' memories. *Better Life* transforms ordinary photographs into better, and even better versions of themselves. We ran multiple tests to check if we would be able to generate this content on the fly, through creative coding platforms such as Processing or OpenFrameworks [34, 44], shown in Figure 9. In our testing, we found these platforms limiting, in that we would have to commit seriously to programming our desired animation. Additionally, after leaving them to run for hours or days, we observed that the frame rate would drop, or the app would crash.

We opted for playing pre-generated videos to avoid this, but also to give us maximum flexibility with changing content up to the last minute of the exhibition opening. However, when playing looped videos, or queuing and triggering videos on-the-fly, we found that performance varied greatly depending on the creative

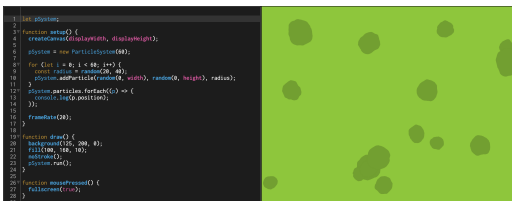


Figure 9: A screenshot of early experiments with p5.js

coding platform we used. We ended up with two systems that gave us satisfactory performance. First, a Raspberry Pi with VLC player controlled through python-vlc, a package that is able to control VLC through python, gave us the simplicity of playing videos from a predefined playlist. Second, an Amazon Fire TV Stick loaded with a custom Unity application that performed more complex transitions, collages, and playback logic, gave us the opportunity to mix different video content with richer animations and effects. Both of these systems gave us high performance because they rely on hardware acceleration to play back video.

Utilising these two ways of video playback, we created the content on different platforms and played it back as video without putting performance strain on the media devices. For HighPoint, videos were generated in TouchDesigner and exported as mp4 files. For Better Life, all photographs were generated with a mixture Generative AI platforms such as DreamStudio [2] and Midjourney [37], then separated into different layers, and layered with pre-generated videos on Unity. For the Protein Machine, we filmed two looped 30-minute videos (one for each aperture of the machine's main compartment), using a variety of materials such as algae, instant coffee, and food colouring to give the illusion of algae growth and fermentation. We then played this on an endless loop with VLC and python-vlc. The result is a looped sequence where seed material is dispensed on the left, and shortly after algae grows and floats on the left, to be sucked into the intestine of the device for further processing.

5.4 Robust physical input mechanisms

All three of the devices require input from the visitor. In HighPoint, different memories can be selected which change the content displayed inside the goggles. In Protein Machine, buttons are used to “extrude” protein, and to browse through the device’s usage logs. In Better Life, the visitor can change the content between Normal, Better, and Even Better, making the subject of the photograph displayed in the frame increasingly farfetched. We suspected that these controls would be used by the public frequently, so we took time to consider what kinds of input components would withstand this level of interaction.

Where we required a rotary interface, we employed encoders, since they are more flexible than potentiometers. If a visitor forces a potentiometer beyond the end of the wiper angle, there is a danger that the case is damaged. Encoders can rotate freely, avoiding this issue entirely. Additionally, most encoders come with an integrated button, which is helpful for selecting content without additional UI. Wherever we required buttons, we utilised high-quality components with higher-than-usual repeat press limits, which were

mounted on panel-mount custom printed circuit boards with components.

Physical inputs must communicate with the software in some way. We debated about how we could achieve a communication between a microcontroller that gathers input events (or drives OLED screens in the case of HighPoint). Early prototypes used a custom serial communication protocol for connecting the physical inputs with software running on a Raspberry Pi. However, this would be trickier to achieve with Amazon Fire TV Stick, which is a more closed system. We instead programmed the microcontrollers to behave as HID devices. This method avoids the use of complex serial communication protocols that we are bound to forget in the future and instead works universally on both Raspberry Pis and the Amazon Fire TV Stick with the addition of a USB OTG hub.

Moving away from custom serial protocols by utilising HID over USB, as well as picking simple and robust inputs such as encoders and push buttons, we ensured that devices could be pressed and interacted with hundreds of times a day.

5.5 Mechanical actuation and geared motion

Whilst designing another prop within the space, a scale model of a drone for the cityscape visible from the room’s window, we started experimenting with simple DC motors. We noticed that not only did the motors make noise, but we also had two fail after two months of testing. We quickly switched to stepper motors, which offered greater flexibility, more precise control, good low-speed torque, and a longer life.



Figure 10: An early test of the helix illusion used in the Protein Machine.

The Protein Machine uses a motor-driven helix to create the illusion of protein extrusion within the transparent chamber. The mechanism combines multiple perceptual effects, including principles similar to the barberpole illusion [49], to suggest a continuous downward stream of protein appearing. Achieving the intended

effect required careful coordination of helix geometry, rotation speed, and enclosure design (very much in the spirit of "smoke and mirrors", discussed in 2.4). To achieve this effect, we utilised a stepper motor, which allowed us to test different speeds without having to change a gearbox or lose torque at lower speeds. An example test configuration is shown in Figure 10. We opted to use a Trinamic stepper driver [3], which provides a variety of current outputs, and has a feature that drives the stepper with high frequency for quiet operation (StealthChop). We utilised StealthChop to reduce vibration on the machine, which can have a detrimental effect to fasteners and mechanisms in the long run [31]. This also influenced our decision to use sound effects to create the illusion of vibration in HighPoint. Instead of using vibration motors often found in EMDR therapy devices [1], we created a series of audible sounds effects that give the idea of vibration, without needing to use any additional moving parts that would require maintenance and would also put mechanical strain on the device over time.

5.6 Physical Access to Components

No matter how robust a device seems, physical access to the hardware is essential for carrying out repairs or updates. All three of our devices include numerous interconnected electronic parts, with a variety of connectors connecting each part, some of which can be seen in the candid photo showing the spread of prototype parts for Better Life and Highpoint (Figure 11). We decided to house electronics outside of the devices themselves, hidden in lockable cupboards underneath each device. This made it easier to update hardware, software, or repair failed components. This resonates with the idea of "breaking the shell", discussed in the related work section.

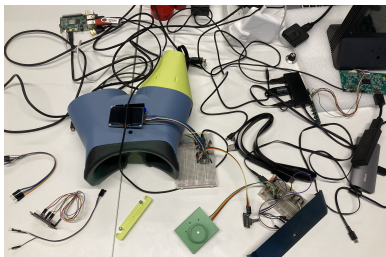


Figure 11: A photo of an early prototype of Highpoint (left) and Better Life (right), showing some of the electronic components spread out over a desk.

5.7 Automatic Startup and Power Cycling

All devices were configured to start up automatically on power up. Fire TV Stick and Raspberry Pi systems were set to boot directly into their respective applications, ensuring that devices recovered from power interruptions, an essential requirement for unsupervised operation in a public exhibition. In addition, we set up timers on the devices so that a daily power cycle is performed. This was intended to enable devices to rest overnight, and to resolve any performance issues that built up during the day.

5.8 Documentation and Source Control

Given the collaborative nature of the project, we relied heavily on documentation and source control using GitHub [25] to help coordinate development. Multiple technologists within the team worked concurrently across the hardware and software of different devices, making the use of GitHub repositories and supporting documentation key for knowledge transfer between.

6 Validating strategies through maintenance

In this section, we discuss the role of maintenance, care and repair when developing the devices for the exhibit. The strategies described in section 5 were informed by prior experience and early prototyping, but their true value could only be assessed through sustained deployment in the museum. In this section, we reflect on how our design decisions held up over the first twelve months of the exhibition. We highlight which strategies proved essential, which required adjustment, and what we failed to anticipate.

6.1 Physical Access Proved Essential

The decision to house electronics in accessible cupboards beneath each device (section 5.6) proved to be invaluable for repair. This was particularly evident in the deployment of Better Life, where we noticed issues every few days during the initial months. Without easy access to the electronics, debugging these issues would have required disassembling the device; a process that would have been both time consuming and disruptive to the exhibition. Instead, we were able to come in, open up the cupboards underneath the devices, and swap components without impacting the exhibition.

6.2 Component Quality and Selection

Selecting components with documented specifications has proven essential for long-term reliability. The encoders in Better Life and Highpoint failed after merely 6 months. When we opened the device to repair it, we saw a ring of debris around the encoder, suggesting that the carbon brushes within it had worn from excessive turning (Figure 12). The original encoders lacked datasheets specifying their rotation lifespan rating, making it impossible to assess their suitability for our high-traffic exhibition. These were since replaced with high-quality encoders manufactured by Bourns and rated for at least 30,000 cycles. These encoders have not shown the same wear in the same time frame as the previous ones.

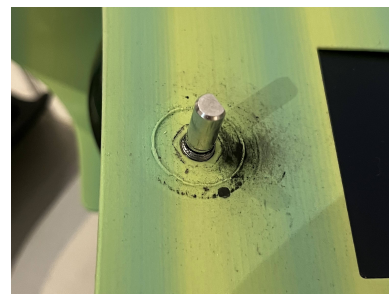


Figure 12: Highpoint's encoder showing debris from excessive use.

We faced similar challenges with the USB hub used to connect the microcontroller to the Amazon Fire TV Stick in Better Life. After a few weeks, we noticed intermittent bugs, with the device unable to read any user input unless power cycled. We identified that the problem was a generic USB hub being incompatible with the official Fire TV Stick power supply. Upgrading this to a higher-specification USB hub with more reliable power regulation resolved the issue.

These experiences underscore the importance of selecting components with available specifications that allow us to verify suitability for the intended application, rather than relying on undocumented, often cheaper, and previously untested parts. Our initial strategy was correct: to use quality components rated for high-traffic use. But our execution fell short.

6.3 Simple Communication Enabled Rapid Debugging

Our decision to use HID-based communication between the electronics and the software (Section 5.5) proved valuable. Because user input signals could be emulated with a standard keyboard, we were able to prototype, debug, and refine the software without requiring the physical hardware to be connected. We were able to test software changes on our own computers before visiting the museum, reducing the number of visits required to resolve issues. The simplicity of the communication protocol also meant that we could quickly isolate whether a problem lay with a microcontroller, the connection, or the software.

6.4 Documentation Enabled Teamwork

The documentation and source control practices described in section 5.8 proved essential for establishing continuity between team members fixing issues that occurred. When responsibility for maintaining or developing a device further was passed between team members, the incoming technologist was able to continue development by referring to each device's GitHub repository and documentation. This approach supports continued maintenance and future development of the devices in the room, as team members join or leave the project over its 10 year lifespan.

6.5 Repairs Carried Out

Over the 12 month period of August 2024 and July 2025, we returned to the museum to carry out repairs, upgrades and content updates 14 times. Table 1 presents a record of our maintenance interventions, categorised into four types: software issues, hardware repairs, content updates, and hardware upgrades. This record reveals that the maintenance activity was concentrated in the early months following the opening of the exhibition, with Highpoint and Better Life requiring the most frequent attention. Hardware repairs, particularly component replacements including encoders and power supplies constituted the majority of interventions.

7 Discussion

The following discussion draws on insights and challenges that emerged during the first year of the exhibition being open to the public. During this time, over 100,000 members of the public visited the room, according to the museum's records. At this point, 12

months into the 10-year lifetime of the exhibition, the three devices have entered a stable phase and now require very little ongoing maintenance.

7.1 Performance and Wear

Aesthetically and materially, the objects now show signs of use, though not always on the devices themselves. The table Highpoint and Better Life are installed on bear visible traces of hundreds of hands: smudges, wear, and scuff marks. These offer a kind of physical patina to the exhibition. The devices however are fabricated from hard-wearing plastics, such as PETG, PLA and PA-12, and have held up well by comparison. The contrast between the worn table surface and the relatively unmarked device enclosures suggests that these material choices were appropriate for the level of interaction the devices receive daily.

That said, several issues and limitations became apparent over time. Due to timers being unplugged by museum staff, devices currently remain powered on continuously, even outside museum hours. This has occasionally led to crashes and software instability, which require a manual power cycle. Incorporating internal timers or power management systems for nightly resets would likely reduce these issues. Additionally, several OLED displays, particularly on Highpoint and Farm Free Protein Machine, are showing early signs of burn-in. This could be mitigated by optimising refresh behaviour or implementing scheduled screen off periods when not in use.

In the Farm Free Protein device, we also encountered performance limitations when introducing new video content. Upgrading from an older model Raspberry Pi to a Raspberry Pi 4 with a larger memory resolved these playback issues and highlighted the importance of hardware headroom when designing for media-heavy installations.

While the devices have remained stable since initial deployment, several uncertainties exist for the years ahead. Key hardware components such as the Raspberry Pi 4 and the Fire TV Stick may be discontinued during the exhibition's lifetime. Our mitigation strategy (simple hardware architecture, documented technical specification, HID interfaces for hardware-to-software communication) is such that it allows substitution of these parts without redesigning the overall system. Additionally, the use of documentation and source control to assist knowledge transfer described in section 5.8 has not truly been tested since the original technologists are still involved in the upkeep of the exhibition.

7.2 Updating Exhibition Content

During the 12-month period, we updated content on several devices, adding new videos and images that help articulate the exhibit's speculative vision of the future. Additional functionality has also been installed into the systems. For example, Highpoint was extended to include handle paddles, enriching the quality and speculative narrative of the piece.

These changes were made feasible by design decisions that prioritised maintenance and accessibility. Without easy access to the core electronics, these iterative updates would likely have required full removal of the devices from the exhibition. While remote software updates over Wi-Fi are often seen as the primary means of

Table 1: Table of fixes in the exhibition

System	Aug '24	Sep '24	Oct '24	Nov '24	Dec '24	Jan '25
Highpoint Memory Intensification System	■ Improve internal cable mgmt	△ Update Video content		★ Upgrade to include new paddle features	△ Update Led patterns	
	■ Replace faulty OLED	■ Replace Power supply		■ Update EMDR Leds	■ Replace Encoder	
	○ Update Controller firmware - UI					
Better Life Picture Frame	○ Software Bug Fix - Animation issue	△ Update Photo content	■ Replace Encoder			
	○ Software Bug - Fix sleep issue					
	△ Update Photo content					
	■ Replace USB hub					
	○ Update Controller firmware - UI					
Farm Free Protein Machine	■ Repair Indicator led					
	■ Replace Raspberry Pi					
System	Feb '25	Mar '25	Apr '25	May '25	Jun '25	Jul '25
Highpoint Memory Intensification System					★ Replace Encoder with Bourns encoder	
Better Life Picture Frame		△ Update Photo content			★ Replace Encoder with Bourns encoder	
					■ Fix loose internal wiring on led indicator	
Farm Free Protein Machine						

Legend: ○ Software Issue ■ Hardware Repair △ Content update ★ Hardware upgrade

maintainability, we found that physical access to the hardware was just as important. It allowed us to replace components, troubleshoot issues, and make adjustments that would not have been possible through software alone. One year later, these considerations, although largely invisible to the visitor, have proven essential to the sustainability of the exhibit.

7.3 Future Improvements and Opportunities

One major unrealised feature was inter-device communication. The architecture was designed to support synchronised behaviours across devices. For example, the system was intended to simulate pandemic lockdowns, energy rationing, or extreme weather events through coordinated content shifts. While the technical foundation for this exists, the feature remains unused. This represents an area of creative potential still to be explored.

It is worth noting that while the wider project involved a larger design team, the technical development, including programming, electronics, hardware integration, and software, was carried out by just two part-time creative technologists. Working within tight time and budget constraints, much of the effort was necessarily focused on ideation and initial deployment, with limited capacity for extended development or feature realisation. These constraints shaped our emphasis on designing robust, maintainable systems

that could be easily updated post-installation without requiring intensive labour.

The first 12 months of opening has reaffirmed that the work of designing tangible, interactive exhibits does not end at installation. Maintenance, iteration, and adaptation are ongoing processes, shaped as much by visitor interaction and environmental context as by the original design intent. These practices reflect a broader ethos of care and responsiveness in the life of technological systems, aligning with the principles of the Long Now. Through following these ideas, longevity and impact are sustained not only by initial robustness of devices, but by a system's capacity to be maintained and re-imagined in context.

7.4 Summary of Strategies

We present a summary of our strategies for developing tangible interactives deployed in public experiences. These have emerged from our development strategies detailed in section 5, as well as our reflections on the success of the deployment after 12 months of the exhibit being open to the public.

Cloudless– Use cloudless systems that do not rely on an internet connection, external servers, or APIs that might change in the future.

Component Quality– Use quality components from reputable suppliers and manufacturers (most importantly) rated at sufficient specifications for long-term deployment.

Video Playback– Opt for video playback for content over generative graphics where possible, to avoid degrading frame rate and unexpected visuals.

Physical Access– Ensure physical access to electronic components is possible, either by taking critical components out of the device, or providing easy access within the case design.

Simple Hardware and Software Communication– Use simple communication protocols between hardware and software, favouring standardised protocols such as HID devices where possible.

Documentation and Source Control– Ensure every aspect of develop has sufficient documentation and source control to aid future repair and extensibility.

Plan Rests– Rest and power cycle devices when not open to visitors to avoid performance lag and allow components to cool down.

Careful Use of Mechanical Movement– Use mechanical movement sparingly. If motors must be used, opt for overspecified brush-free stepper motors.

While these strategies have emerged from a specific museum context, most are not museum-specific. Cloudless architectures, HID-based communication between hardware and software, thorough documentation, and physical access to components for repair are relevant in situations where interactive systems must operate reliably over a long period. Others, such as careful use of mechanical movement and power cycling, apply broadly to any application that requires constant operation and mechanical wear. More context-specific strategies, such as the use of Home Assistant or other cloudless infrastructure for inter-device communication, or the smoke-and-mirrors approach, reflect the needs of this exhibition, but could be adapted for narrative-driven or multiple-device scenarios.

We believe these strategies are relevant for researchers and practitioners creating tangible interactive devices that need to be deployed for a long period in high-footfall public exhibitions.

8 Conclusion and Future Work

This paper offers insights into the challenges faced when developing long term interactive public exhibitions. It highlights considerations that are essential for ensuring longevity, robustness, and adaptability of technological systems over extended periods of deployment. By prioritizing the importance of maintenance and repair in its design, whilst leaving room for future content upgrades and hardware changes, the exhibition can evolve over time without prolonged closures or significant disruptions. These insights gathered contribute a technical design framework, valuable for researchers, designers and technologists undertaking the development of long-standing interactive installations.

The exhibition is programmed to remain open through 2034; plans are underway to expand its speculative vision through updating content on existing devices and integrating new interactive

elements. These strategies in place help keep the exhibition functioning, while allowing for updates to help reinforce the evolving narrative of the future imagined in 2049.

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